

Galois group of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

Since the polynomial $(x^2-2)(x^2-3)$ has four distinct roots $\pm\sqrt{2}, \pm\sqrt{3}$, it is separable.

And, since $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$, $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{3})] \cdot [\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 4$.

So, $|\text{Aut}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 4$ where $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, the splitting field over $\mathbb{Q}$.

Since the automorphism leaving $\mathbb{Q}$ fixed must be conjugate the roots,

there are four possible automorphisms:

$$\begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}.$$

Let $\sigma : K \rightarrow K ; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$

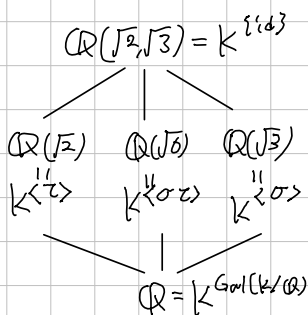
$\tau : K \rightarrow K ; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6}$.

(That is, $\sigma : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$ and $\tau : \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$, and we can observe that

$$\sigma\tau : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \quad \tau\sigma : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \quad \text{so } \sigma^2 = \tau^2 = 1 \text{ and } \sigma\tau = \tau\sigma$$

Now, $\text{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, corresponding σ to $(1, 2)$ and τ to $(2, 1)$.

Moreover,



Note: $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$ because: if $\sqrt{2} = a + b\sqrt{3}$ for some $a, b \in \mathbb{Q}$, then $2 = a^2 + 3b^2 + 2ab\sqrt{3}$.
it is Contradiction

Galois group of $x^3 - 2$ over $\mathbb{Q}$.

Denote $\zeta_3 = e^{i\frac{2\pi}{3}} = \frac{1}{2}(-1 + i\sqrt{3})$, the 3th primitive root of unity.

Since $\zeta_3^3 = 1$, the set $\{\sqrt[3]{2}, \sqrt[3]{2}\zeta_3, \sqrt[3]{2}\zeta_3^2\}$ are all roots of $x^3 - 2$ over $\mathbb{Q}$.
So, the splitting field is $K = \mathbb{Q}(\sqrt[3]{2}, \zeta_3) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{3})$.

Observe that: the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $x^3 - 2$, and
" ζ_3 over $\mathbb{Q}$ is $x^2 + x + 1$.

And, since $x^2 + x + 1$ is irreducible over $\mathbb{Q}(\sqrt[3]{2})$, (it cannot have root in $\mathbb{Q}(\sqrt[3]{2})$)

$$[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}(\sqrt[3]{2})] \cdot [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6.$$

the tower lemma gave. And, since $x^3 - 2$ is separable, $|\text{Aut}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 6$.

Now, construct explicitly the $\text{Aut}(K/\mathbb{Q})$.

Since any automorphism of K leaving $\mathbb{Q}$ fixed must conjugate the roots,

it must map $\sqrt[3]{2}$ to $\sqrt[3]{2}$ or $\sqrt[3]{2}\zeta_3$ or $\sqrt[3]{2}\zeta_3^2$

and maps ζ_3 to ζ_3 or ζ_3^2 .

So, there are exactly 6 possibilities. Consider two automorphisms

σ satisfying $\sigma(\sqrt[3]{2}) = \sqrt[3]{2}\zeta_3$ and $\sigma(\zeta_3) = \zeta_3$

and τ satisfying $\tau(\sqrt[3]{2}) = \sqrt[3]{2}$ and $\tau(\zeta_3) = \zeta_3^2 = -1 - \zeta_3$.

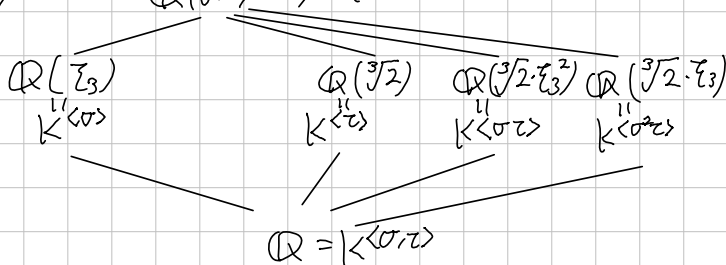
Then, $\sigma^3: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3 \mapsto \sqrt[3]{2}\zeta_3^2 \mapsto \sqrt[3]{2}\zeta_3^3 = \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3 \end{cases}$, $\tau^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \mapsto \zeta_3^4 = 1 - \zeta_3 = \zeta_3 \end{cases}$

are actually identities. Moreover,

$\sigma^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$, $\sigma\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\sigma^2\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\tau\sigma: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$

So, $\text{Aut}(K/\mathbb{Q}) = \{1, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau\} \cong S_3$, corresponding σ to $(1\ 2\ 3)$, τ to $(1\ 2)$

Moreover, $\mathbb{Q}(\sqrt[3]{2}, \zeta_3) = K^{\langle \text{id} \rangle}$



How to find a fixed field corresponding to $\langle \sigma\tau \rangle$?

As we know, the $\text{Tr}_{K/K^{\langle \sigma\tau \rangle}}$ is invariant under any automorphism in $\langle \sigma\tau \rangle$.

So, Calculate:

$$\text{Tr}(\sqrt[3]{2}) = \text{id}(\sqrt[3]{2}) + (\sigma\tau)(\sqrt[3]{2}) = \sqrt[3]{2} + \sqrt[3]{2}\zeta_3 = \sqrt[3]{2}(1 + \zeta_3) = \sqrt[3]{2} \cdot (-\zeta_3^2).$$

Therefore, $\sqrt[3]{2}\zeta_3^2$ is fixed by id and $\sigma\tau$, so $\mathbb{Q}(\sqrt[3]{2}\zeta_3^2) \subseteq K^{\langle \sigma\tau \rangle}$.

Moreover, $[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}(\sqrt[3]{2}\zeta_3^2)] = 2$, so the Galois theorem gives the result.

Galois group of x^8-2 over $\mathbb{Q}$.

By the Eisenstein criterion, $2 \in 2\mathbb{Z}$ but $2 \notin 2^2\mathbb{Z}$, so x^8-2 is irreducible over $\mathbb{Q}$. And, the formal derivative $8 \cdot x^7$ has only root as zero, but $0^8-2 \neq 0$, so the given polynomial is separable. The polynomial x^8-2 has 8 distinct roots,

$$\sqrt[8]{2} \cdot \zeta_8^k \quad (k=0,1,\dots,7) \quad \text{where} \quad \zeta_8 = \exp(2\pi i \cdot \frac{1}{8}) = \frac{\sqrt{2}}{2}(1+i)$$

Therefore, the splitting field $K = \mathbb{Q}(\sqrt[8]{2}, \dots, \sqrt[8]{2} \zeta_8^7) = \mathbb{Q}(\sqrt[8]{2}, i)$, because the $\subseteq$ inclusion is given by $(\sqrt[8]{2})^4 = \sqrt{2}$, so $\zeta_8 \in \mathbb{Q}(\sqrt[8]{2}, i)$, and the $\supseteq$ inclusion is given by $\zeta_8^2 = \frac{1}{2}(2i) = i \in \mathbb{Q}(\sqrt[8]{2}, \dots, \sqrt[8]{2} \cdot \zeta_8^7)$.

By the irreducibility, the minimal polynomial of $\sqrt[8]{2}$ is x^8-2 , and the minimal polynomial of i is x^2+1 trivially. Since $\mathbb{Q}(\sqrt[8]{2}) \subseteq \mathbb{R}$, $i \notin \mathbb{Q}(\sqrt[8]{2})$, thus the tower lemma gives

$$[K:\mathbb{Q}] = [\mathbb{Q}(\sqrt[8]{2}, i):\mathbb{Q}(\sqrt[8]{2})] \cdot [\mathbb{Q}(\sqrt[8]{2}):\mathbb{Q}] = 16.$$

Find explicitly $\text{Gal}(K/\mathbb{Q})$. Since an automorphism in $\text{Gal}(K/\mathbb{Q})$ conjugates the $\sqrt[8]{2}$ and i , therefore, we can define

$$\sigma: \begin{cases} \sqrt[8]{2} \mapsto \sqrt[8]{2} \cdot \zeta_8 \\ i \mapsto i \end{cases} \quad \text{and} \quad \tau: \begin{cases} \sqrt[8]{2} \mapsto \sqrt[8]{2} \quad (\sigma(\zeta_8) = \frac{1+i}{2} \cdot \sigma((\sqrt[8]{2})^4) = \zeta_8^5) \\ i \mapsto -i \quad (\tau(\zeta_8) = \tau(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i = \zeta_8^7) \end{cases}$$

Then, just simple calculation gives $\text{id}, \sigma, \dots, \sigma^7, \tau, \tau\sigma, \dots, \tau\sigma^7$ are mutually distinct. So, $\text{Gal}(K/\mathbb{Q}) = \{\text{id}, \sigma, \dots, \sigma^7, \tau, \dots, \tau\sigma^7\}$. Further, observe that

$$\sigma^8 = \tau^2 = 1 \quad \text{and} \quad \sigma \cdot \tau: \begin{cases} \sqrt[8]{2} \mapsto \sqrt[8]{2} \cdot \zeta_8 \\ i \mapsto -i \end{cases}$$

$$\tau\sigma^3: \begin{cases} \sqrt[8]{2} \xrightarrow{0} \sqrt[8]{2} \cdot \zeta_8 \xrightarrow{1} \sqrt[8]{2} \cdot \zeta_8^6 \xrightarrow{1+5} \sqrt[8]{2} \cdot \zeta_8^7 \xrightarrow{1+30} \sqrt[8]{2} \cdot \zeta_8 \\ i \mapsto -i \end{cases}$$

Therefore $\text{Gal}(K/\mathbb{Q}) = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma \cdot \tau = \tau\sigma^3 \rangle$, the dicyclic group.

$$\text{Gal}(K/\mathbb{Q})$$

$$2 \mid$$

$$\langle \sigma \rangle$$

$$2 \mid$$

$$\langle \sigma^2 \rangle$$

$$2 \mid$$

$$\langle \sigma^4 \rangle$$

$$2 \mid$$

$$\{\text{id}\}$$

Galois group of $x^5 - 3$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity. Observe that: $x^5 - 3$ has five distinct roots, $\sqrt[5]{3} \cdot \zeta^k$ ($k=0,1,\dots,4$).

Since the minimal polynomial of ζ over $\mathbb{Q}$ is $x^4 + x^3 + x^2 + x + 1$, therefore $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$. And, the minimal polynomial of $\sqrt[5]{3}$ over $\mathbb{Q}$ is $x^5 - 3$, therefore $[\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}] = 5$. Therefore, the splitting field of $x^5 - 3$ over $\mathbb{Q}$,

$$K = \mathbb{Q}(\sqrt[5]{3}, \sqrt[5]{3}\zeta, \sqrt[5]{3}\zeta^2, \sqrt[5]{3}\zeta^3, \sqrt[5]{3}\zeta^4) = \mathbb{Q}(\sqrt[5]{3}, \zeta) = \mathbb{Q}(\sqrt[5]{3})\mathbb{Q}(\zeta)$$

has degree of 20 over $\mathbb{Q}$, that is, $[K : \mathbb{Q}] = 20$.

More precisely, by the tower lemma $[K : \mathbb{Q}]$ is divided by $[\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}]$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}]$.

Since 5 and 4 are relatively prime, $[K : \mathbb{Q}] \geq 20$.

And, $x^5 - 3 \in \mathbb{Q}(\zeta)[x]$ has $\sqrt[5]{3}$ as a root, so

$$[\mathbb{Q}(\sqrt[5]{3}, \zeta) : \mathbb{Q}(\zeta)] \cdot [\mathbb{Q}(\zeta) : \mathbb{Q}] \leq 20.$$

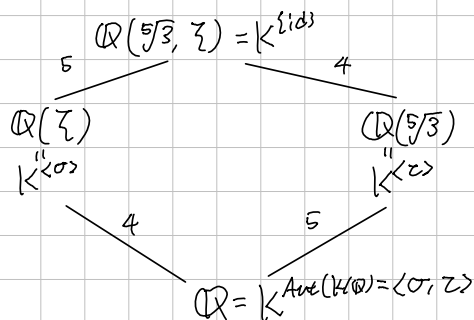
Consequently, $[K : \mathbb{Q}] = 20$. Now, by the Fundamental Theorem of Galois Theory, $K/\mathbb{Q}(\zeta)$ is also Galois, with $[K : \mathbb{Q}(\zeta)] = 5$.

Therefore, $\text{Gal}(K/\mathbb{Q}(\zeta))$ has order of 5, so it must be a cyclic group of order 5. Consequently, the automorphism

$$\sigma : \begin{cases} \sqrt[5]{3} \mapsto \sqrt[5]{3} \cdot \zeta \\ \zeta \mapsto \zeta \end{cases} \quad (\text{It is given by the conjugate isomorphism theorem}).$$

has order of 5, therefore $\text{Gal}(K/\mathbb{Q}(\zeta)) = \langle \sigma \rangle \cong \mathbb{Z}_5$. $\square$

Alternatively,



where $\zeta : \begin{cases} \sqrt[5]{3} \mapsto \sqrt[5]{3} \\ \zeta \mapsto \zeta^2 \end{cases}$ Therefore, $\text{Aut}(K/\mathbb{Q}) \cong \langle (1\ 2\ 3\ 4\ 5), (2\ 3\ 5\ 4) \rangle$.

Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Quid, 2020 Fall)
 For each case, determine the splitting field K and evaluate $[K:F]$.
 And, find the Galois group $\text{Gal}(K/F)$.
 (Note: the field of characteristic 0 or finite field are perfect.)

1) $F = \mathbb{Q}$.

By the Eisenstein criterion, all coefficients of x^6+3 except the leading coefficients are contained in $3\mathbb{Z}$, but the square of constant term $3 \notin 3^2\mathbb{Z}$, the given polynomial x^6+3 is irreducible over $\mathbb{Q}$. The x^6+3 has 6 roots,

$$\sqrt[6]{3} \cdot \omega^{2k+1} \quad (k=0,1,\dots,5) \text{ where } \omega = \exp\left(\frac{i\pi}{6}\right) = \frac{\sqrt{3}}{2} + i\frac{1}{2}.$$

Therefore, the splitting field is $K = \mathbb{Q}(\sqrt[6]{3} \cdot \omega, \sqrt[6]{3} \cdot \omega^3, \dots, \sqrt[6]{3} \cdot \omega^5)$.

Clearly $\mathbb{Q}(\sqrt[6]{3} \cdot \omega) \subseteq K$, by the definition of generated field.

Meanwhile, $(\sqrt[6]{3} \cdot \omega)^3 = \sqrt{3} \cdot \omega^3 = \sqrt{3} \cdot \exp(i\pi/2) = \sqrt{3} \cdot i$ and $\omega^2 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ implies $\omega^2 \in \mathbb{Q}(\sqrt[6]{3} \cdot \omega)$, so $K = \mathbb{Q}(\sqrt[6]{3} \cdot \omega)$. Finally, since x^6+3 is irreducible, $[K:\mathbb{Q}] = 6$.

Moreover, $\text{Gal}(K/\mathbb{Q}) \cong S_3$, because: define two automorphisms on K as:

let $\theta = \sqrt[6]{3} \cdot \omega$ and $\lambda = \omega^2$. Then, $\theta^3 = i\sqrt{3}$ and $\lambda = \frac{1}{2} + \frac{\theta^3}{2}$. Now, let

$$\sigma: \theta \mapsto \theta \lambda \quad \text{and} \quad \tau: \theta \mapsto \theta \lambda^2.$$

Then, $\sigma(\lambda) = \sigma\left(\frac{1}{2} + \frac{1}{2}\theta^3\right) = \frac{1}{2} + \frac{1}{2}\theta^3 \lambda^3 = \frac{1}{2} + \frac{1}{2}i\sqrt{3} \cdot \omega^6 = \frac{1}{2}(1 - i\sqrt{3}) = \omega^{10} = \lambda^{-1}$,
 so $\sigma^2(\theta) = \sigma(\theta \lambda) = \theta \lambda \lambda^{-1} = \theta$, it means $\sigma^2 = \text{id}$.

And $\tau(\lambda) = \frac{1}{2} + \frac{1}{2}\theta^3 \lambda^6 = \frac{1}{2} + \frac{1}{2}i\sqrt{3} = \lambda$, so

$$\tau^2(\theta) = \tau(\theta \lambda^2) = \theta \lambda^4, \quad \tau^3(\theta) = \theta \lambda^6 = \theta, \text{ so } \tau^3 = \text{id}.$$

And, $\sigma\tau(\theta) = \sigma(\theta \lambda^2) = \theta \cdot \lambda^{-1} = \theta \cdot \lambda^2$, $(\sigma\tau)^2(\theta) = \sigma\tau(\theta \lambda^2) = \theta \lambda^{-3} = \theta$,
 therefore $\sigma\tau\sigma\tau = \text{id}$, or $\sigma\tau\sigma = \tau^{-1}$. Now, $\text{Gal}(K/F) = \langle \sigma, \tau \rangle \cong D_6 \cong S_3$.

Alternatively, if $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/6\mathbb{Z}$, the abelian group, then every subgroup is normal. Therefore every intermediate field E satisfies E/F is Galois. (In other say, normal)
 Since $\sqrt[6]{3} \cdot \omega^{11}, \sqrt[6]{3} \cdot \omega = \sqrt[3]{3} \in K$, the field $\mathbb{Q}(\sqrt[3]{3})$ is the intermediate field, but as we know, the minimal polynomial of $\sqrt[3]{3}$ is x^3-3 over $\mathbb{Q}$, but $\mathbb{Q}(\sqrt[3]{3})$ does not have other roots of x^3-3 . So, it is contradiction.

By the knowledge in the group theory, there are only two group of order 6, S_3 and $\mathbb{Z}/6\mathbb{Z}$, so $\text{Gal}(K/\mathbb{Q}) \cong S_3$. □

Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA Qual, 2020 Fall)
 For each case, determine the splitting field K and evaluate $[K:F]$.
 And, find the Galois group $\text{Gal}(K/F)$.
 (Note: the field of characteristic 0 or finite field are perfect.)

2). $F = \mathbb{F}_5$

Suppose that $\alpha \in \overline{\mathbb{F}_5}$ satisfies $\alpha^2=3$. Then, $(\alpha^2)^3+3=3\alpha=0 \pmod{5}$,
 therefore x^2-3 divides x^6+3 , and x^2-3 is irreducible over $\mathbb{F}_5$,
 because substituting $x=0,1,\dots,4$, $x^2-3 \neq 0$, by casework. ($\{x^2 | x \in \mathbb{F}_5\} = \{0,1,4\}$)
 Let $K = \mathbb{F}_5(\alpha)$. Since x^2-3 is the minimal polynomial of α , $[K:\mathbb{F}_5] = 2$.
 Given $a+b\alpha \in K$ ($a,b \in \mathbb{F}_5$), observe that

$$(a+b\alpha)^6 = \underbrace{(a+b\alpha)^5}_{\text{Frobenius automorphism which fixes } \mathbb{F}_5} \cdot (a+b\alpha) = (a+b\alpha^5) \cdot (a+b\alpha) = (a+b\alpha)(a-b\alpha) = a^2 + \underbrace{2b^2}_{=-3} = 2$$

'if and only 'if':

$$\begin{cases} a=0 \\ b=\pm 1 \end{cases} \quad \text{or} \quad \begin{cases} a=\pm 2 \\ b=\pm 2 \end{cases} \quad \Leftrightarrow \quad \begin{cases} a=2 \text{ or } 3 \\ b=2 \text{ or } 3 \end{cases}$$

4 18 22

So, the six distinct roots are contained in K , so K is splitting field of x^6+3 ,
 and clearly it is separable, therefore $K/\mathbb{F}_5$ is Galois, with the Galois group

$$\text{Gal}(K/\mathbb{F}_5) = \langle \alpha \mapsto \alpha^5 \rangle \cong \mathbb{Z}_2.$$

In actually, $K = \mathbb{F}_{5^2}$.

Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA QQual, 2020 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $\text{Gal}(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.)

3). $F = \mathbb{F}_7$

Galois group of $x^p - x + a$ over $\mathbb{F}_p$ where $a \in \mathbb{F}_p \setminus \{0\}$.

First, we need to show that $x^p - x + a$ is irreducible and separable over $\mathbb{F}_p$.

To show the irreducibility, there are two ways:

1). Let $\alpha \in \mathbb{F}_p$ be a root of $x^p - x + a$. That is, $\alpha^p - \alpha + a = 0$.

Notice that $(\alpha+1)^p - (\alpha+1) + a = \alpha^p + 1 - (\alpha+1) + a = 0$, because

on the field of characteristic p , $x \mapsto x^p$ is homomorphism. Inductively,

$\alpha, \alpha+1, \dots, \alpha+p-1$ are roots of $x^p - x + a$. Moreover, these are mutually

distinct, because $\alpha+i = \alpha+j \Leftrightarrow i=j$ but $\{1, \dots, p-1\}$ are distinct in the field of characteristic p . Now, over $\mathbb{F}_p$,

$$x^p - x + a = (x - \alpha)(x - (\alpha+1)) \cdots (x - (\alpha+p-1))$$

If $x^p - x + a$ is reducible over $\mathbb{F}_p$, then $x^p - x + a = f(x)g(x)$ for some $f, g \in \mathbb{F}_p[x]$ and $\deg f, \deg g < p$. By the unique factorization, over $\mathbb{F}_p$,

$$f(x) = \prod_{i \in I} (x - (\alpha+i)) \text{ where } I \neq \{0, 1, \dots, p-1\}.$$

$$\text{But, then } = x^{|I|} - \left(\sum_{i \in I} (\alpha+i) \right) \cdot x^{|I|-1} + \dots,$$

$$\text{so } |I| \cdot \alpha + \sum_{i \in I} i \in \mathbb{F}_p, \text{ but then } |I|^{-1} \cdot |I| \cdot \alpha = \alpha \in \mathbb{F}_p. (0 < |I| < p).$$

But, $\alpha \in \mathbb{F}_p \Leftrightarrow \alpha$ is a root of $x^p - x \Rightarrow a = 0$, it is contradiction.

Therefore, $x^p - x + a$ is irreducible, and separable because all roots are distinct.

Notice that $\mathbb{F}_p(\alpha)$ is the splitting field of $x^p - x + a$ over $\mathbb{F}_p$.

Now, $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] = \deg(x^p - x + a) = p$, therefore

$$\text{Gal}(\mathbb{F}_p(\alpha) / \mathbb{F}_p) = \langle x \mapsto x^p \rangle \cong \mathbb{C}_p, \text{ the cyclic group of order } p$$

Galois group of $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))$ over $\mathbb{Q}$

First, since $x = \sqrt{2+\sqrt{2}} \Rightarrow x^4 - 4x^2 + 2 = 0$, the degree of minimal polynomial of $\sqrt{2+\sqrt{2}}$ divides 4. And, since $1 \notin 2\mathbb{Z}$, $0, 4, 2 \in 2\mathbb{Z}$ but $2 \in 2^2\mathbb{Z}$, so the Eisenstein criterion gives that the given polynomial is irreducible, therefore $x^4 - 4x^2 + 2$ is exactly the minimal polynomial of $\sqrt{2+\sqrt{2}}$.

By the fundamental theorem of Algebra, we can factor $x^4 - 4x^2 + 2$ as linear factors, as

$$\begin{aligned} x^4 - 4x^2 + 2 &= (x^2 - (+2 + \sqrt{2}))(x^2 - (+2 - \sqrt{2})) \\ &= (x - \sqrt{2+\sqrt{2}})(x + \sqrt{2+\sqrt{2}})(x - \sqrt{2-\sqrt{2}})(x + \sqrt{2-\sqrt{2}}) \end{aligned}$$

To show that $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))$ is splitting field of $x^4 - 4x^2 + 2$,

enough to show that $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\sqrt{2+\sqrt{2}})$.

Observe that: $\sqrt{2} = (\sqrt{2+\sqrt{2}})^2 - 2$ and

$$(\sqrt{2+\sqrt{2}}) \cdot (\sqrt{2-\sqrt{2}}) = \sqrt{2}, \text{ so } \sqrt{2-\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2+\sqrt{2}}} = \sqrt{2+\sqrt{2}} - \frac{2}{\sqrt{2+\sqrt{2}}}$$

therefore $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\sqrt{2+\sqrt{2}})$. And, the formal derivative $4x^3 - 8x = 4x(x^2 - 2)$ has roots as $0, \pm\sqrt{2}$, so that $x^4 - 4x^2 + 2$ is separable.

This proves $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))/\mathbb{Q}$ is Galois, and has degree of 4.

Now, consider an automorphism on $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))$ as

$$\sigma: \underset{\alpha''}{\sqrt{2+\sqrt{2}}} \mapsto \underset{\beta}{\sqrt{2-\sqrt{2}}}$$

$$\begin{aligned} \alpha^4 - 4\alpha^2 + 2 &= 0 / \beta = \\ \alpha^3 - 4\alpha &= -\frac{2}{\alpha} \\ -\frac{2}{\alpha} + \beta &= 0 \end{aligned}$$

Then, $\beta = \alpha^3 - 3\alpha$, so

$$\sigma^2(\alpha) = \sigma(\beta) = \sigma(\alpha^3 - 3\alpha) = \sigma(\alpha)^3 - 3\sigma(\alpha) = \beta^3 - 3\beta = \beta - \frac{2}{\beta} = -\alpha$$

So σ has order 4, therefore $\text{Gal}((\mathbb{Q}(\sqrt{2+\sqrt{2}}))/\mathbb{Q}) = \langle \sigma \rangle \cong \mathbb{Z}/4\mathbb{Z}$.

$\mathbb{Q}(\sqrt{2})$ fixed by $\langle \sigma^2 \rangle$

$$\text{Note: } \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = \cos^2 \alpha + \cos^2 \alpha - 1 \Rightarrow \cos^2 \alpha = \frac{1 + \cos 2\alpha}{2} = 1 - \sin^2 \alpha$$

$$\text{So, } \cos \frac{\pi}{8} = \sqrt{\frac{1}{2} + \frac{1}{2} \frac{\sqrt{2}}{2}} = \frac{1}{2} \sqrt{2 + \sqrt{2}} \text{ and } \sin \frac{\pi}{8} = \frac{1}{2} \sqrt{2 - \sqrt{2}}$$

Biquadratic Criterion: $2^2 - 2 = 2$ is not square.

Galois group of $\mathbb{Q}(\sqrt{3+\sqrt{3}})$ over $\mathbb{Q}$

Galois group of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ over $\mathbb{Q}$, where p and q are distinct prime (UCLA string 2023)
 First, $[\mathbb{Q}(\sqrt{p}) : \mathbb{Q}] = 2$ because $x^2 - p$ has $\sqrt{p}$ as root and it is irreducible,
 since $x^2 - p$ has no root in $\mathbb{Q}$.

And, $[\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}(\sqrt{p})] = 2$, because $x^2 - q$ has $\sqrt{q}$ as root and $\sqrt{q} \notin \mathbb{Q}(\sqrt{p})$. Since if $\sqrt{q} \in \mathbb{Q}(\sqrt{p})$, then $\sqrt{q} = a + b\sqrt{p}$, it implies

$$q = a^2 + b^2p + 2ab\sqrt{p} \Rightarrow \sqrt{p} = \frac{a^2 + b^2p}{2ab} \quad (a, b \neq 0)$$

Since $\sqrt{p} \notin \mathbb{Q}$, it is contradiction.

Now, to show $\mathbb{Q}(\sqrt{p}+\sqrt{q}) = \mathbb{Q}(\sqrt{p}, \sqrt{q})$, observe that

$$\sqrt{p} + \sqrt{q} \in \mathbb{Q}(\sqrt{p}, \sqrt{q}) \Rightarrow \mathbb{Q}(\sqrt{p} + \sqrt{q}) \subseteq \mathbb{Q}(\sqrt{p}, \sqrt{q}).$$

By the tower lemma, $[\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}(\sqrt{p})][\mathbb{Q}(\sqrt{p}) : \mathbb{Q}] = 4$,
 enough to show that $[\mathbb{Q}(\sqrt{p} + \sqrt{q}) : \mathbb{Q}] = 4$.

If $[\mathbb{Q}(\sqrt{p} + \sqrt{q}) : \mathbb{Q}] = 1$, then $\sqrt{p} + \sqrt{q} \in \mathbb{Q}(\sqrt{p} + \sqrt{q}) = \mathbb{Q}$, it is contradiction.

If $[\mathbb{Q}(\sqrt{p} + \sqrt{q}) : \mathbb{Q}] = 3$, then

$$4 = [\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}(\sqrt{p} + \sqrt{q})][\mathbb{Q}(\sqrt{p} + \sqrt{q}) : \mathbb{Q}] = n \cdot 3 \text{ for some } n \in \mathbb{N},$$

but trivially $3 \nmid 4$, it contradicts

If $[\mathbb{Q}(\sqrt{p} + \sqrt{q}) : \mathbb{Q}] = 2$, then $\sqrt{p} + \sqrt{q}$ is a root of degree 2 polynomial over $\mathbb{Q}$,
 that is, $(\sqrt{p} + \sqrt{q})^2 + a(\sqrt{p} + \sqrt{q}) + b = 0$ for some $a, b \in \mathbb{Q}$,

$$= \sqrt{pq} + \sqrt{p} + \sqrt{q} + p + q + b = 0 \quad \nexists a, b \text{ such that } \sqrt{pq} + \sqrt{p} + \sqrt{q} + p + q + b = 0$$

$$\Rightarrow \sqrt{pq} = -(\sqrt{p} + \sqrt{q})^2 - (p + q + b) = -(p + q + b)^2 - p - q - 2\sqrt{pq}$$

$$\Rightarrow \sqrt{pq} \in \mathbb{Q}$$

But it is contradiction, so, $\mathbb{Q}(\sqrt{p} + \sqrt{q}) = \mathbb{Q}(\sqrt{p}, \sqrt{q})$.

Clearly, $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ is splitting field of $(x^2 - p)(x^2 - q)$, $\mathbb{Q}(\sqrt{p} + \sqrt{q})/\mathbb{Q}$ is Galois, and define

$$\sigma: \begin{cases} \sqrt{p} \mapsto -\sqrt{p} \\ \sqrt{q} \mapsto \sqrt{q} \end{cases} \quad \text{and} \quad \tau: \begin{cases} \sqrt{p} \mapsto \sqrt{p} \\ \sqrt{q} \mapsto -\sqrt{q} \end{cases}$$

These are well-defined automorphisms of $\mathbb{Q}(\sqrt{p} + \sqrt{q})$ by the Galois theorem.

Moreover, $\text{Gal}(\mathbb{Q}(\sqrt{p} + \sqrt{q})/\mathbb{Q}) = \langle \sigma, \tau \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

②

Galois group of $\mathbb{Q}(\sqrt{p}+\sqrt{q})$ over $\mathbb{Q}$, where p and q are distinct primes, revisited.

First, $[\mathbb{Q}(\sqrt{p}) : \mathbb{Q}] = 2$, because $x^2 - p$ is irreducible over $\mathbb{Q}$, since $x^2 - p$ has no root in $\mathbb{Q}$. And, $[\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}(\sqrt{p})] = 2$, because $x^2 - q$ has no root in $\mathbb{Q}(\sqrt{p})$. Specifically, if $x^2 - q$ has a root in $\mathbb{Q}(\sqrt{p})$, that is, $\sqrt{q} \in \mathbb{Q}(\sqrt{p})$, then $\sqrt{q} = a + b\sqrt{p}$ for some $a, b \in \mathbb{Q}$.

It implies $q = a^2 + b^2p + 2ab\sqrt{p}$,

but since $\sqrt{p} \notin \mathbb{Q}$, either a or b must be zero.

But, $a=0$ means $\sqrt{q} = b\sqrt{p} \Rightarrow q = b^2p$ is not prime when $b \neq 1$, and $q=p$ is contradiction to q and p are distinct, so $b=1$ false.

And, $b=0$ means $\sqrt{q} = a \in \mathbb{Q}$, contradiction. Therefore, by the tower lemma,

$$[\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{p}, \sqrt{q}) : \mathbb{Q}(\sqrt{p})][\mathbb{Q}(\sqrt{p}) : \mathbb{Q}] = 4.$$

To show $\mathbb{Q}(\sqrt{p}+\sqrt{q}) = \mathbb{Q}(\sqrt{p}, \sqrt{q})$, observe that

$$\sqrt{p} + \sqrt{q} \in \mathbb{Q}(\sqrt{p}, \sqrt{q}) \Rightarrow \mathbb{Q}(\sqrt{p} + \sqrt{q}) \subseteq \mathbb{Q}(\sqrt{p}, \sqrt{q})$$

And, observe that

$$p - q = \sqrt{p}^2 - \sqrt{q}^2 = (\sqrt{p} + \sqrt{q})(\sqrt{p} - \sqrt{q}) \Rightarrow \sqrt{p} - \sqrt{q} = \frac{p - q}{\sqrt{p} + \sqrt{q}} \in \mathbb{Q}(\sqrt{p} + \sqrt{q})$$

$$\text{So, } \sqrt{p} = \frac{1}{2}(\sqrt{p} + \sqrt{q} + \sqrt{p} - \sqrt{q}) \text{ and } \sqrt{q} = \frac{1}{2}(\sqrt{p} + \sqrt{q} - (\sqrt{p} - \sqrt{q}))$$

are contained in $\mathbb{Q}(\sqrt{p} + \sqrt{q})$. Thus $\mathbb{Q}(\sqrt{p} + \sqrt{q}) = \mathbb{Q}(\sqrt{p}, \sqrt{q})$.

And, $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ is the splitting field of $(x^2 - p)(x^2 - q)$ over $\mathbb{Q}$.

Therefore, $\mathbb{Q}(\sqrt{p}, \sqrt{q})/\mathbb{Q}$ is Galois, so it has four automorphism that fixes $\mathbb{Q}$.

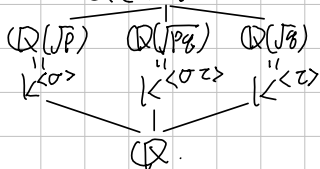
Now, we can explicit two automorphisms:

$$\sigma: \begin{cases} \sqrt{p} \mapsto \sqrt{p} \\ \sqrt{q} \mapsto -\sqrt{q} \end{cases} \text{ and } \tau: \begin{cases} \sqrt{p} \mapsto -\sqrt{p} \\ \sqrt{q} \mapsto \sqrt{q} \end{cases}$$

Then, $\sigma^2 = \tau^2 = (\sigma\tau)^2 = \text{id}$ by just simple calculation.

Consequently, $\text{Gal}(\mathbb{Q}(\sqrt{p}, \sqrt{q})/\mathbb{Q}) = \langle \sigma, \tau \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$,

$$\mathbb{Q}(\sqrt{p} + \sqrt{q}) = K = \mathbb{Q}(\sqrt{p}, \sqrt{q})$$



Galois group of $\mathbb{Q}(x)/\mathbb{Q}(x^2)$

Γ Galois group of $\mathbb{Q}(x,y)/\mathbb{Q}(xy)$

